European Journal of Applied Physiology
https://doi.org/10.1007/s00421-024-05536-w

ORIGINAL ARTICLE

# Biceps femoris long head stiffens after 2 weeks of training cessation in highly trained sprinters

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Received: 2 May 2024 / Accepted: 13 June 2024
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## Abstract

**Purpose** Most athletes experience short-term training cessation because of illness, injury, post-season vacation, or other reasons. Passive muscle stiffness is a potential risk factor for a sprint-type hamstring strain injury, but limited information is available about the effect of short-term training cessation on passive muscle stiffness. The present study aimed to identify whether and how passive muscle stiffness of the biceps femoris long head (BFlh) would vary due to 2 weeks of training cessation in sprinters.

**Methods** Passive BFlh shear-wave speed (a proxy for stiffness) was measured using ultrasound shear-wave elastography in 28 male sprinters, before and after 2 weeks of intervention. During the 2 weeks, the participants in the training-cessation group ($n = 14$) were allowed to maintain their normal daily activities but not to perform any physical training, including stretching and resistance exercises. The participants in the training continuation group ($n = 14$) performed the training (including maximum speed sprint, plyometric, and weight training) prescribed by their coaches 5 days per week.

**Results** In the training-cessation group, passive BFlh shear-wave speed increased after the 2 weeks of training cessation ($4.75 \pm 0.77$ to $5.00 \pm 0.88$ m/s, $P < 0.001$). In contrast, there was no significant difference before and after the 2 weeks of training continuation ($4.90 \pm 0.85$ to $4.93 \pm 0.85$ m/s, $P = 0.521$).

**Conclusions** The present findings indicate that muscles stiffen by training cessation in sprinting athletes.

**Keywords** Hamstring · Stiffness · Shear-wave speed · Elastography · Detraining

**Abbreviations**

| | |
|---|---|
| ANCOVA | Analysis of covariance |
| ANOVA | Analysis of variance |
| BFlh | Biceps femoris long head |
| HSI | Hamstring strain injury |
| SWE | Shear-wave elastography |
| PKE | Passive knee extension |

## Introduction

Most athletes experience short-term (days to a few weeks) training cessation due to injury, illness, post-season vacation, or other reasons. Although short-term training cessation could optimize athletes' performance via physiologic and/or psychologic recovery (Bosquet et al. 2013; Travis et al. 2022), anecdotal reports indicate that power athletes such as track sprinters and soccer players are prone to suffer from sprint-type hamstring strain injuries (HSI) when they return to training or play after such periods of training

Communicated by Michael I Lindinger.

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Published online: 21 June 2024

stoppage. The cause of HSI is multifactorial, including both non-modifiable (e.g., age, injury history) and modifiable factors (e.g., reduced hamstring strength, stiff hamstring) (Buckthorpe et al. 2019; Green et al. 2020; Timmins et al. 2016). Regarding the latter factors, to date, attention has been paid to the effects of short-term training cessation on muscle strength in athletes. In contrast, limited information is available on the effect of short-term training cessation on hamstring stiffness in athletes. Evidence on short-term training cessation-induced changes in hamstring stiffness can lead to an advanced understanding of HSI risk factors and help clinicians and practitioners perform effective treatment for injury prevention and rehabilitation.

Previously, hamstring stiffness has been evaluated using joint torque–angle curves of passive hip flexion (Marshall et al. 2009) or knee extension (Magnusson et al. 1997). However, the evaluation of these tests is influenced by not only the hamstring stiffness but also non-muscular tissues (e.g., tendons, ligaments). Additionally, these tests cannot distinguish the stiffness between three constituent muscles of the hamstring (i.e., the biceps femoris long head (BFlh), semitendinosus, and semimembranosus), and thus it is extremely challenging to identify muscle-specific characteristics and their changes. Alternatively, ultrasound shear-wave elastography (SWE) has been recently used to evaluate the stiffness of individual muscles independently of the influences of subjectivity and muscle size (Eby et al. 2013; Hirata et al. 2018; Maïsetti et al. 2012; Miyamoto et al. 2018b). Briefly, SWE generates shear waves using focused pushing beams (acoustic radiation force) within a tissue and measures the shear-wave propagation speed using ultrafast ultrasound imaging(Bercoff et al. 2004; Palmeri et al. 2008). Shear-wave speed can be used as a proxy for the stiffness of the tissue because of its relation to the elastic moduli, such as shear modulus and Young's modulus; the stiffer the tissue, the faster the shear wave propagates (Eby et al. 2013). There is a growing body of evidence indicating that passive muscle stiffness is strongly affected by intramuscular connective tissues (such as perimysium and endomysium) (Ducomps et al. 2003; Gajdosik 2001; Gillies and Lieber 2011). More recently, high passive muscle stiffness assessed with ultrasound SWE has been suggested as a novel potential risk factor related to muscle-strain injuries (Kumagai et al. 2019; Miyamoto-Mikami et al. 2021). Furthermore, although there is no consensus regarding the characteristics of passive muscle stiffness in sprinting athletes (Avrillon et al. 2020; Miyamoto et al. 2019), passive muscle stiffness in sprinters has been reported to be lower than that in untrained individuals (Miyamoto et al. 2019). Passive muscle stiffness has been reported to be reduced by longitudinal training, such as plyometric (Ando et al. 2021) and stretching exercises (Ichihashi et al. 2016) that sprinters commonly perform. Therefore, it is reasonable to assume that passive muscle stiffness of sprinters increases by stopping their training. When considering these observations, together with the fact that most sprint-type HSI occurs in the BFlh (Askling et al. 2007; Koulouris et al. 2007) and anecdotal report that athletes are prone to suffer from sprint-type HSI when returning to training or play after short-term training cessation, we can hypothesize that passive BFlh stiffness would increase after short-term training cessation. We aimed to examine whether and how passive BFlh stiffness would vary due to 2 weeks of training cessation in sprinting athletes.

## Methods

### Participants

Twenty-eight healthy male track-and-field sprinters (20.0 ± 1.2 years, 173.7 ± 5.6 cm, 65.6 ± 6.6 kg) participated in the present study. The inclusion criteria were (i) current participation in competitive races and (ii) at least three years of regular training in track-and-field sprinting. The exclusion criterion included any current lower extremity injuries or illness that would prevent them from regular training before 2 weeks of intervention. They were allocated to either the training-cessation group ($n = 14$) or the training-continuation group ($n = 14$) to match their physical characteristics, specialized disciplines, and performance level (Table 1). In both groups, eight participants specialized in 100-m, two in 200-m, and four in 400-m. Before participation, all participants were fully informed of the purpose and experimental procedures and gave their written consent to participate. This study was approved by the institutional ethics committee (No. 2021-055) and performed in accordance with the Declaration of Helsinki.

**Table 1** Physical and performance profiles of training-cessation and training-continuation groups

| | Training Cessation ($n = 14$) | Training Continuation ($n = 14$) |
|---|---|---|
| Age (years) | 20.4 ± 1.3 | 19.4 ± 0.9 |
| Height (cm) | 174.1 ± 4.3 | 173.3 ± 6.5 |
| Weight (kg) | 65.6 ± 4.0 | 65.5 ± 8.5 |
| World Athletics points for season best time | 984.4 ± 109.0 | 961.2 ± 88.0 |

## Experimental protocol and setting

Measurements of passive BFlh shear-wave speed (a proxy for stiffness) were conducted 1 day before and just after 2 weeks of intervention. During the 2-week intervention period, the participants in the training-cessation group were allowed to maintain their normal daily activities but not to perform any physical training, including stretching and resistance exercises. The 2 weeks of training cessation started immediately after the last competition of the season. All the participants in both groups had a normal in-season period, which included regular training prescribed by their coaches 5 days per week (weekly training hours: approximately 15 h in average), until the last competition. The training programs leading to the last competition were similar in structure, but not identical across participants; that is, the details of the training programs (e.g., exercises, loads, repetitions, and sets) depended on the participants. The participants in the training-cessation group stopped training on the same day. The participants in the training-continuation group continued their usual coach-prescribed training programs (including maximum speed sprint, resisted sled sprint, uphill sprint, sprint technique drills, plyometric, and weight training) 5 days per week during the 2 weeks of intervention. Pre- and post-intervention measurements were performed at the same time of day for each participant. The participants were not allowed to perform strenuous exercises for 24 h and warm-ups or stretching exercises for 1 hour before the measurements.

The procedure for measuring muscle shear-wave speed and the participants' posture were similar to those used in previous studies (Kumagai et al. 2019; Miyamoto et al. 2018a; Miyamoto-Mikami et al. 2021). The shear-wave speed of the BFlh of the dominant leg was assessed using an ultrasound SWE scanner (Aixplorer Ver. 12, Supersonic Imagine, France) in 'SWE' mode (preset: 'MSK', persistence: off, smoothing: 5). The dominant leg was defined as the preferred jumping leg. The participants were seated on a bench with their hip flexed at 70° (0° = lying position), their knees fully extended, and their ankles at approximately 90° (Fig. 1). This hip joint angle was chosen according to previous studies (Kumagai et al. 2019; Miyamoto et al. 2018a; Miyamoto-Mikami et al. 2021) so that the hamstring could be stretched to a tensioned state without pain for all participants and that the muscle shear-wave speed could be assessed at a certain angle common to all participants. An ultrasound probe (SL10-2, Supersonic Imagine, France) was positioned at 50% of the thigh length (defined as the distance between the lateral epicondyle of the femur and the greater trochanter). The probe orientation was manually aligned with the BFlh fascicle direction so that several fascicles were identified within the B-mode image in a certain plane. In the post-intervention measurements, the probe orientation was further adjusted while referring to the several landmarks (such as fat, connective tissues, and blood vessels) in the B-mode images acquired before the intervention, according to previous studies (Ema et al. 2013; Hirata et al. 2017). Care was taken not to deform the muscles while scanning. The participants were instructed to relax the leg fully throughout the measurements. The images were acquired after ensuring a stable color distribution of the elastogram window for a few seconds. The measurements were performed three times. All measurements were performed by two experienced examiners with more than 7 years of practice. The reliability of measuring passive BFlh stiffness using ultrasound shear-wave elastography by the examiners was previously confirmed, with an intraclass correlation coefficients of 0.99 (Miyamoto et al. 2017). The room temperature for all measurements was kept constant at approximately 25 °C to minimize any potential temperature-induced effects on tissue stiffness. The SWE images were exported to a DICOM format, and image processing was conducted using a custom-written MATLAB algorithm. The algorithm calculated the mean shear-wave speed within the region of interest that was as large as possible with the exclusion of other tissues (e.g., subcutaneous fat and aponeurosis). Intramuscular connective tissues such as the perimysium and endomysium were included. Only the pixels with a quality index of more

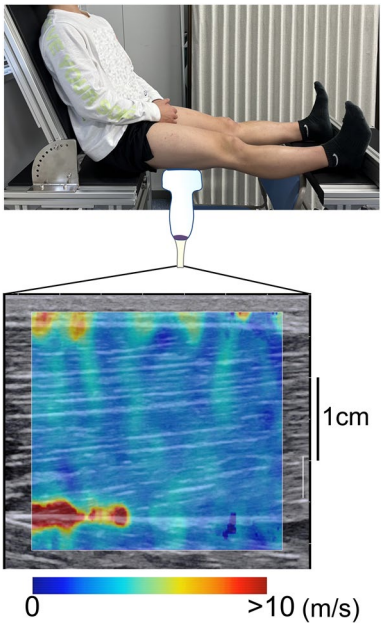


**Fig. 1** Experimental setup for ultrasound shear-wave elastography (SWE) measurement. The bottom image is a typical example of SWE measurement. The colored region represents the shear modulus map with the scale below the image

than 0.7 (out of the maximum of 1.0) were included in the analyses (Chodock et al. 2020). We confirmed that no pixel in the region of interest reached the saturation limit of shear-wave speed for the SWE system (16.3 m/s). The average value of three measurements was used for subsequent analyses.

### Statistical analysis

A priori power analysis with an assumed type 1 error of 0.05 and statistical power of 0.80 was conducted to find a significant interaction of passive muscle stiffness between group and time, using G*Power 3.1.9.4 (Kiel University, Germany). Regarding the effect size, there were no direct data from previous studies to be referred to because no previous study has examined the effects of training cessation on muscle stiffness. However, it is likely that the magnitude of the effect of training cessation is influenced by that of the training effect. Thus, we used the effect size of 0.46 based on a longitudinal intervention study that examined the effect of plyometric training that sprinters commonly perform and showed a significant group × time interaction of passive muscle stiffness assessed using ultrasound SWE (Ando et al. 2021). The minimum sample size per group was estimated to be 10. Therefore, the 14 participants per group in the present study satisfied the minimum sample size.

The normality of passive BFlh shear-wave speed was confirmed using the Shapiro–Wilk normality test. An unpaired t test was conducted on the pre-intervention value to assess whether a significant baseline difference existed between groups. A two-way repeated measures analysis of variance (ANOVA) (within-subject factor: time [before vs. after], between-subject factor: group [training cessation vs. training continuation]) was used to compare passive BFlh shear-wave speed. When a significant interaction was found, post-hoc paired *t* tests were performed to determine whether significant differences existed before and after the intervention. Additionally, the primary comparison of intervention effects was a between-group comparison to the intervention. Therefore, the data were also analyzed using analysis of covariance (ANCOVA) with the corresponding pre-intervention values used as covariates. The statistical analyses were performed using statistical software (SPSS Statistics Ver. 29, IBM Japan, Japan).

## Results

Figure 2 shows the BFlh shear-wave speed before and after the 2 weeks of intervention in the training-cessation and training-continuation groups. At baseline, no significant difference was observed between groups for the passive BFlh shear-wave speed ($P = 0.606$). Two-way ANOVA showed a significant group × time interaction ($P < 0.001$, effect size of partial $\eta^2 = 0.399$). Follow-up paired *t* tests revealed that the BFlh shear-wave speed was significantly greater after than before the intervention in the training-cessation group (before: 4.75 ± 0.77 m/s, after: 5.00 ± 0.88 m/s, $P < 0.001$, effect size of Cohen's $d = 0.305$) but not in the training-continuation group (before: 4.90 ± 0.85 m/s, after: 4.93 ± 0.85 m/s, $P = 0.521$, effect size of Cohen's $d = 0.026$). ANCOVA demonstrated that after adjusting for the nonsignificant pre-intervention difference, the BFlh shear-wave speed at the post-intervention in the training-cessation group was significantly greater than that in the training-continuation group ($P < 0.001$).

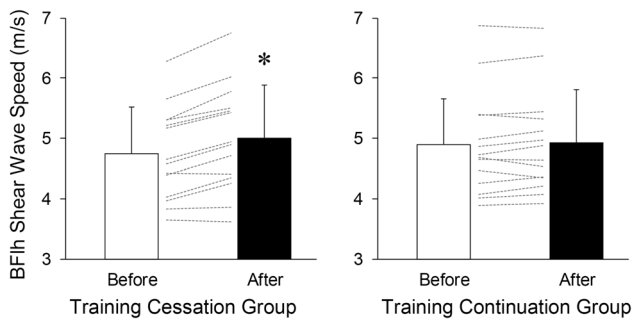


**Fig. 2** Shear-wave speed of biceps femoris long head (BFlh) before and after 2 weeks of intervention in training-cessation and -continuation groups. *Significantly different from before the intervention ($P < 0.05$)

## Discussion

To the best of our knowledge, this is the first study to demonstrate the effect of short-term training cessation on passive muscle stiffness. The novel finding is that passive BFlh shear-wave speed increased after the 2 weeks of training cessation, whereas it remained unchanged after the 2 weeks of training continuation. This finding indicates that muscles stiffen by training cessation as short as 2 weeks in sprinting athletes.

Recently, most studies with ultrasound SWE have examined the impacts of various types of exercise on passive muscle stiffness (Hirata et al. 2016, 2020; Ichihashi et al. 2016; Kawama et al. 2022; Miyamoto et al. 2017, 2020; Morales-Artacho et al. 2017; Zhi et al. 2022). Specifically, passive muscle stiffness of the hamstring constituents, including the BFlh, was reported to decrease after both an acute bout (Miyamoto et al. 2020) and chronic training (Ichihashi et al. 2016) of stretching exercise. More recent studies have demonstrated that passive muscle stiffness is also acutely reduced by eccentric contractions, i.e., active lengthening exercises (Kawama et al. 2022; Zhi et al. 2022)

and stretch-shortening cycle exercise training (Ando et al. 2021). Among several possible mechanisms for the passive and active lengthening-induced reduction in passive muscle stiffness, intramuscular connective tissue lengthening is the most likely and plausible explanation (Gajdosik 2001; Gillies and Lieber 2011) because passive muscle stiffness is strongly affected by intramuscular connective tissues such as the perimysium and endomysium. The intramuscular connective tissues have wavy fibrils, which are commonly referred to as crimp (Diamant et al. 1972). The crimps within the connective tissues disappeared after lengthening the tissues (Franchi et al. 2007a). Furthermore, such crimps are suggested to decrease in number and appear more flattened after the tissue is physiologically elongated to an extent in vivo (Franchi et al. 2007a, 2007b). It is reasonable to presume that these phenomena would occur during both passive lengthening and active lengthening (i.e., eccentric exercise). Collectively, we can conclude that skeletal muscle is a tissue susceptible to the effects of daily/habitual exercise in terms of its mechanical properties. In general, athletes (including sprinters as recruited in the present study) often perform stretching, eccentric (e.g., Nordic hamstring exercise), and/or stretch–shortening cycle exercises (e.g., sprinting, plyometrics) for performance enhancement and injury prevention. Indeed, passive muscle stiffness of sprinters is lower than that of untrained individuals (Miyamoto et al. 2019). Taken together, passive muscle stiffness of sprinters would be increased by stopping such their regular training. Namely, the increased passive muscle stiffness after the 2 weeks of training cessation in the present study is likely to be attributable to intramuscular connective tissue adaptation.

The present findings may have practical implications for preventing muscle injuries due to increased muscle stiffness and improving muscle conditioning for superior performance. Regarding the former, although the precise risk factors and mechanisms for sprint-type HSI are unclear, passive muscle stiffness assessed with ultrasound SWE has been recently suggested as a potential risk factor related to muscle-strain injuries (Kumagai et al. 2019; Miyamoto-Mikami et al. 2021). On the other hand, there are anecdotal reports from coaches indicating that sprinting athletes are prone to suffer from sprint-type HSI when they return to training or play after short-term (days to a few weeks) training stoppage. Considering these anecdotal reports together with the present finding, although an increase in passive muscle stiffness after the intervention does not directly lead to an increased risk of injury if the muscle stiffness is within the normal range, sprinting athletes and their coaches should focus on recovering muscle stiffness after the training cessation. For the latter implication, high passive muscle stiffness has been recently reported to be beneficial to jump height (Ema 2022) and rapid explosive production of muscle force and power (Ando and Suzuki 2019; Ema 2022; Yamazaki et al. 2022). Given these findings, the present findings suggest that explosive muscular performance (e.g., rate of force/torque development, power, jump performance) is likely improved after short-term training cessation. Nevertheless, the present study did not directly assess the association of passive BFlh stiffness or its change due to training cessation with HSI incidence and muscular performance. Thus, further longitudinal investigations are warranted to conclude whether passive BFlh stiffness is related to the sprint-type HIS susceptibility or sport-performance enhancement.

A limitation of the present study is that participating participants were restricted to males. Passive muscle stiffness assessed using ultrasound SWE is reportedly lower in female than male athletes (Kumagai et al. 2019; Miyamoto et al. 2018b; Miyamoto-Mikami et al. 2021), probably due to the effect of sex hormones such as estrogen (Miyamoto et al. 2018b). Thus, it is possible that the effect of short-term training cessation on passive muscle stiffness varies between male and female athletes, i.e., it is unclear whether the current findings hold for female athletes. Another limitation is that passive BFlh stiffness was assessed only on the dominant leg for all participants. However, a previous study using ultrasound SWE showed that the laterality did not affect the passive BFlh stiffness in elite athletes of symmetric (such as track-and-field sprinters) and asymmetric sport activities (such as taekwondo and fencing) (Avrillon et al. 2020). Thus, it is unlikely that there are dominant/non-dominant differences in passive BFlh stiffness and their changes due to the training cessation. Third, we did not control or assess a change in intramuscular water content due to a difference in muscle glycogen storage before and after the intervention, although it is unclear whether such changes affect muscle shear-wave speed. However, the changes in intramuscular water content and muscle glycogen storage are also the effect of training cessation. Finally, the timing of the measurements was limited to 2 weeks after training cessation. Thus, further research is needed to determine when passive muscle stiffness increases after training cessation and whether passive muscle stiffness increases further in the case of training cessation longer than 2 weeks.

In conclusion, we provided evidence that passive muscle stiffness is increased after 2 weeks of training cessation in sprinters, whereas passive muscle stiffness remains unchanged when they continue training.

**Author contributions** Conceptualization: NM, KY, TI, IM, DY, KH; Methodology: NM, DY, KH; Formal analysis and investigation: NM, DY, KH; Project administration: NM, KY, TI; Supervision: NM, IM; Writing—original draft: NM, KH; Writing—review and editing:

KY, TI, IM, DY; Funding acquisition: DY. All authors have read and approved the manuscript.

**Funding** This study was part of the "Functional Development Project for Resilient Athlete Support", a project commissioned by Japan Sports Agency.

**Data availability** The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

## Declarations

**Conflict of interest** The authors have no conflicts of interest to declare that are relevant to the content of this article.

**Ethical approval** This study was approved by the ethics committee (No. 2021–055) of Japan Institute of Sports Sciences. The study procedures adhered to the tenets of the Declaration of Helsinki.

**Consent to participate** The procedure, purpose, and risks associated with this study were explained to the participants, and written informed consent was obtained.

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